

USE CASE TABLES

Author	<i>Elda Toska</i>
UC Name	<i>UC-01&04 Search/Explore Products</i>
Summary	<i>The system allows customers to search, browse, filter, and sort products using keywords, categories, subcategories, and various criteria.</i>
Dependency	• <i>Depends on product catalog and search engine.</i> Integrates with • <i>UC-05 – Compare Products (Elda Toska)</i> • <i>UC-06 – Search by Image (Elda Toska)</i> • <i>UC-25 –View Personalized Recommendations,</i>
Actors	<i>Primary: Customer</i>
Preconditions	- <i>Product catalog exists</i> - <i>Customer has access to the system</i>
Description of the Main Sequence	• <i>Customer enters keywords in the search bar.</i> • <i>Customer browses categories or subcategories.</i> • <i>Customer applies filters (price, rating, availability, etc.) .</i> • <i>Customer selects sorting criteria (popularity, recency, etc.).</i> • <i>System processes the request.</i> • <i>System retrieves matching products.</i> • <i>System displays product listings.</i>
Description of the Alternative Sequence	• <i>- No keywords entered → system shows default or popular products.</i> • <i>- No filters applied → system shows broader results.</i> • <i>- No matching products → system displays “No results found”.</i>
Non-functional requirements	<i>The system shall display product pages and search results within an acceptable response time.It should handle large product datasets efficiently.</i>
Postconditions	<i>Customer is shown a list of relevant products</i>

Author	<i>Elda Toska</i>
UC Name	<i>UC-02 – View Product Availability</i>
Summary	<i>The system allows customers to view product availability status (in stock / out of stock).</i>
Dependency	• <i>Depends on product catalog and inventory data.</i>

	• Depends on product catalog and inventory data. • Depends on: • UC-01 – Explore/Search Products (Elda Toska) Integrates with: • UC-05 – Compare Products (Elda Toska) • UC-25 – View Personalized Recommendations, • FR-36 – Seller monitors stock levels
Actors	Primary: Customer
Preconditions	- Product catalog exists - Inventory data is available - Customer has access to the system
Description of the Main Sequence	• Customer views a product listing or product details page. • System retrieves product availability status. • System displays whether the product is in stock or out of stock.
Description of the Alternative Sequence	• Inventory data unavailable → system displays “Availability unknown”.
Non-functional requirements	The system shall display availability status in real time without noticeable delay.
Postconditions	Customer is informed about product availability status.

Author	Elda Toska
UC Name	UC-03 – Share Product Listings
Summary	The system allows customers to share product listings via external platforms such as social media or direct links.
Dependency	• Depends on product catalog and product page availability. Depends on: • UC-01 – Explore/Search Products (Elda Toska) Integrates with: • UC-05 – Compare Products (Elda Toska)

	UC-06 – Search by Image (Elda Toska)
Actors	Primary: Customer
Preconditions	- Product listings are available - Customer has access to the system
Description of the Main Sequence	- Customer selects a product. - Customer clicks the “Share” option. - System generates a shareable link or sharing options. - Customer selects a platform or copies the link. - System completes the sharing action.
Description of the Alternative Sequence	External platform unavailable → system displays error message..
Non-functional requirements	The system shall generate shareable links instantly and ensure compatibility with major external platforms.
Postconditions	Product listing is shared or a shareable link is generated.

Author	Elda Toska
UC Name	UC-05 Compare Products
Summary	The system allows customers to compare multiple products based on price, specifications, ratings, and availability.
Dependency	Depends on product database and product attribute data. Depends on: UC-01 – Search/Explore Products
Actors	Primary: Customer
Preconditions	Multiple products are available Customer has selected at least two products
Description of the Main Sequence	- Customer selects multiple products - Customer initiates the comparison - System retrieves product details - System compares product price - System compares product specifications - System compares product ratings - System compares product availability

	• System displays comparison results
Description of the Alternative Sequence	• Less than two products selected → system prompts user to select more • Product data unavailable → system shows partial comparison
Non-functional requirements	<i>The comparison must be fast and responsive. Data should be accurate and clearly presented. The system should handle multiple product comparisons efficiently.</i>
Postconditions	<i>Customer is presented with a detailed product comparison.</i>

Author	<i>Elda Toska</i>
UC Name	<i>UC-06 Search by Image</i>
Summary	<i>The system allows customers to upload an image and find visually similar products.</i>
Dependency	<i>Depends on image processing system and product catalog.</i> Depends on: • UC-01 – Search/Explore Products
Actors	<i>Primary: Customer</i>
Preconditions	<i>Image upload feature is available</i> <i>Product catalog exists</i>
Description of the Main Sequence	• Customer initiates image search • Customer uploads an image • System processes the image • System searches for similar products • System retrieves matching products • System displays results
Description of the Alternative Sequence	• Invalid image format → system displays error message • No similar products found → system shows alternative suggestions
Non-functional requirements	<i>The system must process images efficiently with low latency. It should ensure accuracy in matching products and handle large image datasets.</i>
Postconditions	<i>Customer is presented with visually similar products</i>

Author	<i>Elda Toska</i>
UC Name	<i>UC-07 View Recently Viewed Products</i>
Summary	<i>The system allows customers to view a list of products they have recently viewed.</i>
Dependency	<i>Depends on user session data or browsing history storage.</i> Depends on: • <i>UC-01 – Search/Explore Products</i> Integrates with: • <i>UC-25 – View Personalized Recommendations</i>
Actors	<i>Primary: Customer</i>
Preconditions	<i>Customer has previously viewed products</i> <i>System stores browsing history</i>
Description of the Main Sequence	• <i>Customer navigates to recently viewed section</i> • <i>System retrieves browsing history</i> • <i>System displays the list of recently viewed products</i> • <i>Customer selects a product (optional)</i>
Description of the Alternative Sequence	• <i>No browsing history available → system displays empty message</i> • <i>Customer does not select any product → use case ends</i>
Non-functional requirements	<i>The system must retrieve data quickly and ensure user privacy. It should efficiently store and manage browsing history.</i>
Postconditions	<i>Customer views their recently viewed products.</i>

Author	<i>Sindi Terihati</i>
UC Name	<i>UC-08 Shopping Cart Management</i>
Summary	<i>The customer manages the shopping cart by adding products, updating product quantities, and removing products before checkout.</i>
Dependency	Depends on: • <i>UC-01&04 Search/Explore Products</i> Integrates with:

	• UC-10 – Order and Checkout Management • UC-20 – Reorder Products
Actors	• Primary: Customer
Preconditions	• Customer is browsing the platform • Product exists in the catalog
Description of the Main Sequence	• Step 1: Customer selects a product • Step 2: Customer adds the product to the shopping cart • Step 3: System checks product availability • Step 4: System stores the product in the cart • Step 5: Customer opens the shopping cart • Step 6: Customer updates the quantity of a selected product • Step 7: System recalculates the cart total • Step 8: Customer removes a product from the cart • Step 9: System updates the shopping cart contents
Description of the Alternative Sequence	• 2a. Product is out of stock → system displays an unavailable message • 6a. Invalid quantity entered → system requests a valid quantity • 8a. Last item removed → system displays an empty cart message: ...
Non-functional requirements	The system must provide a clear and intuitive shopping cart interface so customers can easily add, update, and remove products. It should respond quickly to cart changes and update totals without noticeable delay. The shopping cart must also be compatible with major web browsers and mobile devices to ensure a consistent user experience across platforms.
Postconditions	• Shopping cart contents are updated • Cart total reflects the current selected products and quantities

Author	Sindi Terihati
UC Name	UC-09 Wishlist Management
Summary	The customer saves products to a wishlist for future purchases.
Dependency	Depends on: • UC-01&04 Search/Explore Products

	<i>Integrates with:</i> • <i>UC-17 – Notifications (Elisa Spaho)</i> • <i>UC-25 – View Personalized Recommendations</i>
Actors	• <i>Primary: Customer</i>
Preconditions	• <i>Product exists in the catalog</i> • <i>Customer is logged into an account</i>
Description of the Main Sequence	• <i>Step 1: Customer select a product</i> • <i>Step 2: Customer chooses to add the product to the wishlist</i> • <i>Step 3: System saves the product in the customer's wishlist</i> • <i>Step 4: Customer opens the wishlist</i> • <i>Step 5: System displays all saved wishlist items</i>
Description of the Alternative Sequence	• <i>2a. Customer is not logged in → system requests login or account creation</i> • <i>3a. Product already exists in wishlist → system notifies customer</i> • <i>5a. Saved product is unavailable → system marks the product as unavailable</i>
Non-functional requirements	<i>The system must provide an easy-to-use wishlist interface that allows customers to save products with minimal effort. It should synchronize wishlist data reliably so saved items remain available across sessions. The functionality must also work properly across major browsers and mobile devices.</i>
Postconditions	• <i>Selected product is saved in the wishlist</i> • <i>Wishlist displays the customer's saved items for future reference</i>

Author	<i>Sindi Terihati</i>
UC Name	<i>UC-10 Order and Checkout Management</i>

Summary	<i>The customer places orders, completes purchases using available payment methods, and cancels orders before shipment.</i>
Dependency	Depends on: • UC-08 – Shopping Cart Management Integrates with: • UC-46-47 – Order Validation & Invoice Generation (Arjola Metaliaj) • UC-23 – Manage Delivery Addresses (Stilianos Troka) • UC-22 – Manage Payment Methods (Stilianos Troka)
Actors	• Primary: Customer • Secondary: Payment Gateway
Preconditions	• Customer is logged into an account • Shopping cart contains at least one product • Selected products are available in stock
Description of the Main Sequence	• Step 1: Customer opens the shopping cart • Step 2: Customer proceeds to checkout • Step 3: System displays order summary, payment methods, and delivery details • Step 4: Customer selects a payment method • Step 5: Customer confirms the purchase • Step 6: Payment system processes the transaction • Step 7: System creates the order • Step 8: System displays purchase confirmation
Description of the Alternative Sequence	• 4a. Payment method is invalid → system requests another payment method • 6a. Payment fails → system displays transaction failure message • 7a. Product becomes unavailable during checkout → system prevents order completion • 8a. Customer requests cancellation before shipment – System verifies shipment status – System cancels the order and updates the order status – Order has already been shipped → cancellation is rejected and customer is notified
Non-functional requirements	<i>The system must ensure reliable transaction processing and preserve order records in case of failure through backup and recovery mechanisms. It should provide a clear checkout interface that allows customers to complete purchases efficiently with minimal confusion. The checkout process must also perform consistently across major browsers and mobile devices.</i>
Postconditions	• Order is successfully created if payment is completed • Order status is updated if cancellation occurs before shipment • Transaction record is stored in the system

Author	<i>Sindi Terihati</i>
UC Name	<i>UC-11 Guest Checkout</i>
Summary	<i>The customer completes a purchase without creating an account by using guest checkout.</i>
Dependency	<i>Depends on:</i> • <i>UC-08 – Shopping Cart</i> <i>Integrates with:</i> • <i>UC-46-47 – Order Validation & Invoice Generation</i>
Actors	• <i>Primary: Customer</i> • <i>Secondary: Payment Gateway</i>
Preconditions	• <i>Shopping cart contains at least one product</i> • <i>Selected products are available for purchase</i>
Description of the Main Sequence	• <i>Step 1: Customer opens the shopping cart</i> • <i>Step 2: Customer proceeds to checkout</i> • <i>Step 3: System offers the option to sign in or continue as guest</i> • <i>Step 4: Customer selects guest checkout</i> • <i>Step 5: System requests shipping, billing, and payment information</i> • <i>Step 6: Customer enters the required information</i> • <i>Step 7: Customer confirms the purchase</i> • <i>Step 8: Payment system processes the transaction</i> • <i>Step 9: System creates the guest order</i> • <i>Step 10: System displays order confirmation</i>
Description of the Alternative Sequence	• <i>4a. Customer changes decision → system redirects to login or registration</i> • <i>5a. Required information is incomplete → system requests missing fields</i> • <i>8a. Payment fails → system displays an error and asks for another attempt</i>
Non-functional requirements	<i>The guest checkout process must be simple and efficient so customers can complete purchases quickly without registration. The system must handle guest transaction data reliably and support recovery in case of service interruption. It should also remain accessible and functional across major browsers and mobile devices.</i>
Postconditions	• <i>Guest order is recorded in the system</i> • <i>Customer completes purchase without creating an account</i> • <i>Order confirmation is displayed</i>

Author	<i>Sindi Terihati</i>
UC Name	<i>UC-12 Interface Theme Switching</i>
Summary	<i>The customer switches between light mode and dark mode for the user interface.</i>
Dependency	<i>Independent use case. No dependency on shopping or checkout functionalities.</i> Depends on: • <i>None</i> Integrates with: • <i>UC-13 – Language Selection</i>
Actors	• <i>Primary: Customer</i>
Preconditions	• <i>Customer is using the platform interface</i>
Description of the Main Sequence	• <i>Step 1: Customer opens interface settings or theme controls</i> • <i>Step 2: Customer selects light mode or dark mode</i> • <i>Step 3: System applies the selected theme</i> • <i>Step 4: System updates the interface appearance</i> • <i>Step 5: System saves the preference for future sessions if supported</i>
Description of the Alternative Sequence	• <i>2a. Customer is not logged in → system applies theme only for the current session</i> • <i>3a. Theme cannot be loaded properly → system keeps default theme and shows a message</i>
Non-functional requirements	<i>The system must provide a responsive and user-friendly interface for theme switching without disrupting normal platform usage. Theme changes should be applied immediately and consistently across supported browsers and devices. The interface must remain readable and visually consistent in both light and dark modes.</i>
Postconditions	• <i>Selected theme is applied successfully</i> • <i>Interface reflects the customer's preferred appearance setting</i>

Author	<i>Sindi Terihati</i>
UC Name	<i>UC-13 Language Preference Selection</i>
Summary	<i>The customer selects a preferred language for the user interface.</i>
Dependency	<i>Independent use case. No dependency on cart, wishlist, or checkout functionalities.</i> Depends on: • <i>None</i> Integrates with: • <i>UC-12 – Theme Switching</i>
Actors	• <i>Primary: Customer</i>
Preconditions	• <i>Customer is using the platform interface</i> • <i>At least one language option is available in the system</i>
Description of the Main Sequence	• <i>Step 1: Customer opens language settings</i> • <i>Step 2: Customer selects a preferred language</i> • <i>Step 3: System applies the selected language to the interface</i> • <i>Step 4: System refreshes or updates visible text elements</i> • <i>Step 5: System saves the language preference for future sessions if supported</i>
Description of the Alternative Sequence	• <i>2a. Selected language is temporarily unavailable → system keeps the current language and shows a notification</i> • <i>4a. Some interface content is not translated → system displays default language content for those elements</i>
Non-functional requirements	<i>The language selection feature must be easy to access and simple to use for customers. The system should apply language changes quickly and consistently across interface pages without requiring complex actions. It must also maintain compatibility across major browsers and mobile devices while preserving interface clarity and usability.</i>
Postconditions	• <i>Selected language is applied to the interface.</i> • <i>Customer interacts with the platform in the preferred language</i>

Author	Elisa Spaho
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UC Name	UC-15 – Submit Refund Requests and Monitor Refund Status
Summary	Enables a customer to submit a refund request for an eligible purchase, monitor its status throughout the resolution lifecycle, and receive an automatic refund when an order is cancelled before shipment. (Based on: FR-15)
Dependency	Depends on: • UC-10 – Order and Checkout Management (<i>Sindi Terihati</i>) • UC-19 – View Complete Order History (<i>Elisa Spaho</i>) Integrates with: • UC-17 – Receive Notifications. (<i>Elisa Spaho</i>) • UC-39-40 – 'Vendor Subscription & Dashboard Access' (<i>Asja Xhelezi</i>) • UC-29 – Submit Support Ticket (<i>Ederta Kariqi</i>)
Actors	Primary Actor: Customer Secondary Actor: Administrator
Preconditions	• The customer is authenticated and logged into their SOSS account. • The customer has at least one order with status 'Delivered' or 'Completed' in their order history • The order falls within the platform's defined refund eligibility period. • Payment for the order has been confirmed.
Description of the Main Sequence	1. Customer navigates to 'My Orders' from the account dashboard. 2. Customer selects the completed or delivered order for which a refund is required. 3. Customer clicks 'Request Refund' on the order detail page. 4. System verifies refund eligibility based on order status, purchase date, and product type. 5. Customer selects a refund reason from a predefined list and optionally adds comments and supporting evidence. 6. Customer reviews the request summary and submits. 7. System creates the refund record with a unique reference number and status 'Pending Review'. 8. System sends a confirmation notification via the customer's preferred channel. 9. Customer monitors live refund status (Pending Review → Approved / Rejected → Refund Issued) from the order detail page.

Description of the Alternative Sequence	Alt-A — Order not eligible for refund: – System determines the order does not meet eligibility criteria and displays the applicable policy reason. – System directs the customer to submit a support ticket. Alt-B — Refund rejected by administrator: – Administrator rejects the request with a stated reason. – System updates status to 'Rejected' and notifies the customer. Alt-C — Automatic refund from order cancellation: – Customer cancels an order before shipment. – System automatically initiates a refund to the original payment method without requiring a manual request. – Customer receives cancellation and refund confirmation notifications.
Non Functional Requirements	The system must ensure high reliability and financial accuracy during refund processing by maintaining consistent transaction and order data across all modules within five minutes, with zero tolerance for discrepancies. All refund state changes, transactions, and user actions must be fully logged to support auditing and regulatory compliance. In case of inconsistencies, the system shall automatically trigger a reconciliation process and notify administrators promptly. Additionally, all data handling must comply with applicable data protection regulations.
Postconditions	• A refund request record is created with a unique reference number and status 'Pending Review' . • The customer receives a confirmation notification with the reference number and expected timeline. • Refund status is visible and accessible from order history at all times. • Upon administrator approval, the system initiates the refund to the customer's original payment method.

Author	Elisa Spaho
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UC Name	UC-16 – Track Real-Time Delivery via Live Map and View Estimated Delivery Times
Summary	Allows a customer to track the real-time location of their delivery via live map updates and view estimated delivery times based on shipping method and delivery location. Based on: FR-16
Dependency	Depends on: • <i>UC-10 – Order and Checkout Management (Sindi Terihati)</i> • <i>UC-48-49 – Platform Analytics & Inventory Synchronisation (Arjola Metaliaj)</i> • <i>UC-23 – Manage Delivery Addresses (Stilianos Troka)</i> Integrates with: • <i>UC-44-45 – Product Information Display (Arjola Metaliaj)</i> • <i>UC-17 – Receive Notifications (Elisa Spaho)</i> • <i>UC-NEW-3 – Guest Order Tracking and Invoice Access (Sindi Terihati)</i>
Actors	Primary Actor: Customer Secondary Actor: Logistics Provider – external system supplying real-time tracking data via the API integration
Preconditions	• The customer is authenticated and logged into their SOSS account. • The customer has an active order with status 'Shipped' or 'Out for Delivery'. • The logistics provider has assigned a tracking number to the order. • The system's logistics API integration is operational.

Description of the Main Sequence	1. Customer navigates to 'My Orders' and selects an active order with 'Shipped' or 'Out for Delivery' status. 2. Customer clicks 'Track My Delivery' on the order detail page. 3. System authenticates and queries the external logistics API using the order's tracking number. 4. System receives real-time GPS coordinates and shipment status data from the logistics API. 5. System renders an interactive map displaying the current delivery location and destination pin, fulfilling. 6. System calculates and displays the Estimated Time of Arrival based on current location, route, and shipping method. 7. Map and ETA refresh automatically every 30 seconds without page reload. 8. Order detail page displays a shipment status timeline with timestamps for each completed stage. 9. When the logistics provider marks the order 'Delivered', the system updates the order status and sends a delivery confirmation notification.
Description of the Alternative Sequence	Alt-A — Logistics API temporarily unavailable: – System fails to connect to the logistics API after two retry attempts. Error logged per NFR-12. – System displays the last successfully retrieved tracking status with its timestamp. – A notice informs the customer that live tracking is temporarily unavailable. Core checkout and order functions remain operational per NFR-22. Alt-B — Tracking number not yet active: – Logistics API returns 'Tracking not yet available'. – System displays the estimated delivery date from FR-45 as a fallback and informs the customer that live tracking will be available within a few hours. Alt-C — Guest customer access: – A guest customer enters their order reference number and email address. – System authenticates the request and provides tracking access without requiring account login.
Non Functional Requirements	The system must provide efficient and reliable real-time delivery tracking by maintaining a synchronization latency of no more than 30 seconds with external logistics providers and automatically reconnecting in case of connection failures. It must support multiple logistics APIs without requiring code changes and ensure detailed logging of any failed external interactions. The tracking functionality must be accessible across mobile and desktop platforms and support graceful degradation so that core system operations remain unaffected if tracking services fail.

Postconditions	• The customer sees the current real-time delivery location on an interactive live map. • The estimated time of arrival is displayed and updated dynamically. • The shipment status timeline reflects the most recent delivery stage. • Upon delivery confirmation, the order status updates to 'Delivered' and the customer is notified.
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Author	Elisa Spaho
UC Name	UC-17 – Receive Notifications for Order Updates, Delivery Status, Price Drops, Restocking and Promotional Offers
Summary	Allows the customer to receive notifications for order updates, cancellation, delivery status changes, price drops, promotional offers and product restocking. Based on: FR-17 . Also covers system-level.
Dependency	Depends on: • <i>UC-10 – Order and Checkout Management (Sindi Terihati)</i> <i>UC-09 – Wishlist Management (Sindi Terihati)</i> • <i>UC-21 – Manage Privacy Settings (Elisa Spaho)</i> Integrates with: • <i>UC-46-47 – Order Validation & Invoice Generation (Arjola Metaliaj)</i> • <i>UC-41-42 – Customer Engagement Analytics (Asja Xhelezi)</i> • <i>UC-48-49 – Platform Analytics & Inventory Synchronisation (Arjola Metaliaj)</i> • <i>UC-36 – Monitor Inventory Levels (Asja Xhelezi)</i>
Actors	Primary Actor: Customer Secondary Actor: Logistics Provider – provides delivery status events (FR-49)

Preconditions	• The customer has an active SOSS account. • The customer has not disabled the relevant notification category in their preferences. • A triggering event has occurred: order placed , shipment status changed, price drop detected, or wishlist product restocked.
Description of the Main Sequence	1. A notification-triggering event occurs: order placed / cancelled, order shipped or delivered , price drop , or promotional offer. 2. System notification engine detects the event and identifies the relevant customers. 3. System checks each customer's notification preferences. 4. System generates a contextually appropriate notification message including relevant details. 5. System dispatches the notification via all enabled channels (email, push, in-app). Order status notifications are delivered within 30 seconds. 6. Notification is delivered 7. Notification is logged in the customer's in-app notification centre. 8. Customer opens the notification and is deep-linked to the relevant platform section.
Description of the Alternative Sequence	Alt-A — Customer has disabled a notification category: – System suppresses delivery for that category per customer's preferences. – Event is still logged in the in-app notification centre as a record. Alt-B — Notification delivery failure: – Dispatch fails for one channel. Failure logged. – System retries on the failed channel once after 2 minutes, then attempts any other enabled channel. Alt-C — Price drop sub-flow: – System detects a price decrease on a product saved in a customer's wishlist. – System identifies all customers with the product in their active wishlist. – Each customer receives a price drop notification showing the original and new price. Alt-D — Restock notification sub-flow: – A seller updates a previously out-of-stock product. – System identifies customers who opted in to 'Notify me when back in stock'. – Each opted-in customer receives a restock notification with a direct link to the product page.

Non Functional Requirements	The system must deliver timely and reliable notifications, ensuring that real-time updates such as order status changes are sent within 30 seconds, while price drop and restock alerts are delivered within five minutes. All notification events must be logged for auditing purposes, and any failures in notification delivery must generate detailed error logs. The system must also comply with data protection regulations and ensure that all notifications respect user consent and privacy preferences.
Postconditions	• Customer has received a notification via their enabled channel(s) for the triggered event. • Notification is recorded in the in-app notification centre with event details and timestamp. • For price drop notifications, the customer is directed to the product page showing the updated price. • For order and delivery notifications, the customer is directed to the order tracking or order details on page.

Author	Elisa Spaho
UC Name	UC-18 – Download Auto-Generated Invoices or Receipts for Each Completed Purchase
Summary	Allows a customer to download an auto-generated invoice or receipt for each completed purchase. Invoices are generated automatically by the system upon order completion and are accessible from the order history at any time. Based on: FR-18,FR-46.
Dependency	Depends on: • UC-10 – Order and Checkout Management (Sindi Terihati) • UC-46-47 – Order Validation & Invoice Generation (Arjola Metaliaj) • UC-19 – View Complete Order History (Elisa Spaho) Integrates with: • UC-17 – Receive Notifications (Elisa Spaho) • UC-29 – Submit Support Ticket (Ederta Kariqi) • UC-NEW-3 – Guest Order Tracking and Invoice Access (Sindi Terihati)
Actors	Primary Actor: Customer

Preconditions	• The customer is authenticated and logged into their SOSS account, or is a guest with an order reference number and verified email. • The customer has at least one completed order. • The system has already generated the invoice.
Description of the Main Sequence	1. Customer navigates to 'My Orders' and selects a completed order. 2. Customer clicks 'Download Invoice' or 'Download Receipt' on the order detail page. 3. System retrieves the pre-generated PDF invoice from secure storage. 4. System initiates the file download to the customer's device. 5. PDF is saved to the customer's device with a filename including the order reference number and date. Invoice retained.
Description of the Alternative Sequence	Alt-A — Invoice not yet generated: – System determines the invoice has not yet been generated. – System displays: 'Your invoice is being prepared and will be available shortly.' – Once generated, the customer receives an in-app notification. Alt-B — PDF retrieval failure: – System cannot retrieve the PDF. Error logged. – System displays a meaningful error message and a 'Try Again' option. – If the issue persists, customer is directed to submit a support ticket. Alt-C — Guest customer access: – Guest customer enters order reference number and email address. – System authenticates the request and provides invoice download access without requiring account login.
Non Functional Requirements	The system must ensure secure and reliable access to invoices by protecting sensitive financial data from unauthorized access and logging every access event for auditing purposes. Invoice data must remain fully consistent with the corresponding transaction records, and all records must be retained according to regulatory requirements. In case of errors, the system must provide clear and user-friendly messages without exposing internal system details.

Postconditions	• Customer has successfully downloaded a PDF invoice or receipt. • The downloaded file accurately reflects all transaction details for the selected order. • The invoice remains stored and downloadable from order history.
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Author	<i>Elisa Spaho</i>
UC Name	UC-19 – View Complete Order History Including Past Purchases, Order Status and Transaction Details
Summary	<i>Allows a customer to view their complete order history including past purchases, order status and transaction details. The order history is the central hub for all post-purchase actions including tracking, refunds, invoices, and reordering. Based on: FR-19 – Elisa Spaho</i>
Dependency	Depends on: • UC-10 – Order and Checkout Management (Sindi Terihati) Integrates with: • UC-15 – Submit Refund Requests (Elisa Spaho) • UC-16 – Track Delivery (Elisa Spaho) • UC-18 – Download Invoice (Elisa Spaho) • UC-20 – Reorder from History (Elisa Spaho). • UC-NEW-1 – View Delivery Carbon Footprint (Elisa Spaho)
Actors	Primary Actor: Customer
Preconditions	• The customer is authenticated and logged into their SOSS account. • The customer has placed at least one order on the platform.
Description of the Main Sequence	1. Customer navigates to 'My Orders' from the account dashboard or profile menu. 2. System retrieves all orders associated with the customer's account. 3. System displays a paginated, chronologically ordered list of orders showing: order reference number, order date, total amount paid, number of items, current order status, and a product thumbnail. 4. Customer optionally filters the list by order status (All, Active, Delivered, Cancelled) or searches by order reference number or product name. 5. Customer selects an order to view its full detail page. System displays the complete order detail including: itemised products with quantities and unit prices; payment method used; delivery address ; applied promotions; subtotal, shipping cost, taxes and total paid; order status timeline

	<i>with timestamps; and available actions: Track Delivery , Download Invoice, Request Refund , Reorder , View Carbon Footprint</i>
Description of the Alternative Sequence	Alt-A — No order history exists: – System displays: 'You have not placed any orders yet.' – System presents a 'Start Shopping' call-to-action linking to product browsing. Alt-B — Search or filter returns no results: – System displays: 'No orders match your search criteria' and suggests clearing the filter. Alt-C — System fails to load order history: – System displays a user-friendly error message and a 'Retry' button. Error logged per. – If the error persists, customer is directed to submit a support ticket.
Non-functional requirements	<i>The system must maintain accurate and consistent order history records, ensuring alignment with transaction data across all modules within five minutes. Access to order history must be securely restricted to authenticated users, with all access logged for auditing purposes. The system must retain order history for a minimum of three years and provide a clear, intuitive, and responsive interface accessible across both mobile and desktop devices.</i>
Postconditions	• The customer can view a complete, accurate, and up-to-date list of all their orders. • The customer can navigate to any individual order's full detail view. <i>The customer can initiate further actions from the order detail page: Track Delivery, Download Invoice, Request Refund, Reorder, View Carbon Footprint.</i>

Author	Elisa Spaho
UC Name	UC-20 – Reorder Previously Purchased Products Directly from Order History
Summary	Allows a customer to reorder previously purchased products directly from their order history, with live stock verification and transparent communication of any unavailability or price changes. Based on: FR-20

Dependen cy	Depends on: • UC-19 – View Complete Order History (Elisa Spaho) • UC-08 – Shopping Cart Management (Sindi Terihati) • UC-36 – Monitor Inventory Levels (Asja Xhelezi) Integrates with: • UC-10 – Order and Checkout Management (Sindi Terihati) • UC-43 – Personalised Product Recommendation (Arjola Metaliaj)
Actors	Primary Actor: Customer
Preconditi ons	• The customer is authenticated and logged into their SOSS account. • The customer has at least one previous order in their history. • The customer's shopping cart is accessible and may already contain items.
Descriptio n of the Main Sequence	1. Customer navigates to 'My Orders' and locates the previous order to repeat. 2. Customer clicks the 'Reorder' button on the order list entry or order detail page. 3. System performs live stock checks on each product from the original order (inventory maintained by : Seller shall monitor stock levels). 4. System adds all available products to the customer's current shopping cart with original quantities. 5. Reordered items are appended to the cart without removing any existing contents. 6. System displays a reorder summary listing items added, items unavailable, and any price changes since the original order. 7. Customer reviews the cart and proceeds to checkout or continues shopping.

Description of the Alternative Sequence	Alt-A — All items unavailable: – No items are added. System displays: 'None of the items from this order are currently available.' – System suggests similar products via the recommendation engine. Alt-B — Some items unavailable: – Available items are added to the cart. Unavailable items are listed in the summary with their status. – Customer can proceed to checkout with available items. Alt-C — Price has changed since original order: – Items are added but the summary clearly shows the original and current price for each changed item. – Customer can remove individual items from the cart if they no longer wish to purchase at the new price.
Non Functional Requirements	The system must support a seamless and user-friendly reorder process by ensuring that product, inventory, and order data remain consistent across all modules. The user interface must be intuitive, with the reorder option clearly visible and easily accessible. In case of failures, the system must provide meaningful error messages without exposing internal technical details.
Postconditions	• All available products from the selected order are added to the cart with their original quantities. • Customer has been informed of any unavailable items and any price changes. • The cart is ready for checkout.

Author	Elisa Spaho
UC Name	UC-21 – Manage Privacy Settings and Configure Per-Event Notification Preferences

Summary	Allows a customer to manage privacy settings to control how their personal data and browsing activity are used for recommendations and marketing purposes, including configuring notification preferences individually for each event type. Based on: FR-21 .
Dependency	Depends on: <i>None. This use case can be accessed independently at any time from Account Settings.</i> Integrates with: • UC-25 – View Personalised Product Recommendations (Stilianos Troka) • UC-43 – Personalised Product Recommendation (Arjola Metaliaj) • UC-17 – Receive Notifications (Elisa Spaho) • UC-NEW-2 – Request Personal Data Export
Actors	Primary Actor: Customer
Preconditions	• The customer is authenticated and logged into their registered SOSS account. • The customer has navigated to Account Settings.
Description of the Main Sequence	1. Customer navigates to 'Account Settings' and selects the 'Privacy & Data' section. 2. System displays current privacy preferences in labelled categories: Personalised Recommendations on/off, Browsing History Tracking on/off, Marketing Communications on/off, per-event Notification Preferences, and Data Sharing for Analytics on/off. 3. System also displays a plain-language summary of what data is currently collected and how it is used. 4. Customer toggles individual settings on or off: Customer shall manage privacy settings including configuring notification preferences individually for each event type. 5. Customer clicks 'Save Preferences'. 6. System validates and saves the updated preferences. Every change is timestamped and logged. 7. System immediately applies the new settings, sending suppression signals to the recommendation engine and notification engine as applicable. 8. System displays confirmation: 'Your privacy preferences have been updated successfully.'

Description of the Alternative Sequence	Alt-A — Customer disables all data usage: – System displays a notice explaining the functional impact: personalised recommendations, targeted notifications, and marketing communications will be disabled. – Customer confirms. System applies all suppression settings . – Customer continues with generic non-personalised content. Alt-B — Save fails due to a system error: – System retains previous settings, displays an error message and error is logged. – Customer is prompted to try again or submit a support ticket. Alt-C — Customer requests personal data export: – Customer selects 'Download My Data' within Privacy & Data settings. – System creates a data export request and processes it within 72 hours (GDPR Article 20 compliance). – Customer receives an email confirmation of the export request with a reference number.
Non Functional Requirements	The system must ensure secure handling of user privacy settings by enforcing role-based access control, allowing only authenticated users to modify their own preferences. All changes must be logged with timestamps and previous values to support auditing. The system must comply with data protection regulations and ensure that all personal data is securely stored and protected from unauthorized access.
Postconditions	• Customer's privacy preferences are updated and saved. • Settings are propagated to the recommendation engine and notification engine immediately. • A timestamped audit log entry is created for compliance. • Customer receives a confirmation message that preferences have been saved. • If personalised recommendations are disabled, the system falls back to popular products.

Author	<i>Elisa Spaho</i>
UC Name	UC-NEW-1 – View Actual Carbon Footprint of Completed Delivery Within Order History
Summary	<i>Allows a customer to view the actual carbon footprint generated by their completed delivery within their order history, based on the shipment route and method used. This closes the sustainability loop opened by FR-30 (Ederta Kariqi), which shows an estimate at checkout — this use case shows the actual post-delivery figure. Based on: FR-NEW-1</i>
Dependency	Depends on: • UC-19 – View Complete Order History (Elisa Spaho). • UC-10 – Order and Checkout Management (Sindi Terihati) • UC-48-49 – Platform Analytics & Inventory Synchronisation (Arjola Metaliaj) Integrates with: - UC-30 – View Carbon Footprint Report (Ederta Kariqi)
Actors	Primary Actor: Customer
Preconditions	• The customer is authenticated and logged into their SOSS account. • The customer has at least one order with status 'Delivered'. <i>The system has received actual shipment route data from the logistics provider for the completed order.</i>
Description of the Main Sequence	1. Customer navigates to 'My Orders' (UC-19) and selects a completed and delivered order. 2. Customer scrolls to the 'Sustainability' or 'Delivery Impact' section of the order detail page. 3. System retrieves the actual shipment route and method data from the logistics API integration. 4. System calculates the actual carbon footprint based on the real delivery route, vehicle type, and distance covered. 5. System displays the actual carbon footprint figure in kg CO₂ equivalent, alongside a comparison to the estimate shown at checkout. 6. System displays a brief contextual note explaining what the figure means (e.g. equivalent to X km driven by an average car).
Description of the Alternative Sequence	Alt-A — Actual route data not yet available: – System displays the pre-purchase estimate as a fallback with a note: 'Actual delivery footprint will be available once shipment data is confirmed.' Alt-B — Carbon footprint data unavailable from logistics provider: – System displays: 'Carbon footprint data is not available for this delivery.' – The pre-purchase estimate remains visible if it was recorded at checkout.
Non-functional	<i>The system must provide accurate carbon footprint information by</i>

requirements	<i>synchronizing with external logistics data within defined performance limits and ensuring consistency with order records. This feature must not impact core system functionality and should support graceful degradation if unavailable. Any errors must be communicated clearly to users without exposing internal system details.</i>
Postconditions	<i>• The customer has viewed the actual carbon footprint of their completed delivery within the order detail page.</i> <i>• The figure displayed reflects the real shipment route and method, not the pre-purchase estimate.</i> <i>If actual data is unavailable, the pre-purchase estimate is shown as a fallback with appropriate labelling.</i>

Author	<u>Stilianos Troka</u>
UC Name	UC-26 Export Personal Data
Summary	<i>This use case allows the customer to request and download an export of their personal data, including account information, order history, and related activity, in a structured and downloadable format.</i>
Dependency	Includes: • User Authentication and Identity Verification • Data Aggregation Service • Export File Generation Service Depends on: • UC-21 – Manage Privacy Settings • Customer Account/Profile Management • Order History Management • GDPR Compliance and Privacy Management Module Integrates with: • UC-17 – Receive Notifications • UC-19 – View Complete Order History • UC-29 – Submit Support Ticket • Audit Logging and Compliance Tracking System
Actors	<i>Primary Actor: Customer</i>
Preconditions	• Customer is registered in the system. • Customer is logged into their account. • Customer has personal data stored in the system (e.g., orders, profile data).
Description of the Main Sequence	• Step 1: Customer navigates to the account/privacy settings section. • Step 2: Customer selects the “Export Personal Data” option. • Step 3: System displays export options (e.g., data scope, format). • Step 4: Customer requests the data export. • Step 5: System validates the request and confirms authorization. • Step 6: System collects customer data (account details, order history, etc.). • Step 7: System generates the export file in a structured format (e.g., JSON, PDF, CSV). • Step 8: System prepares the file for download. • Step 9: System notifies the customer that the export is ready. • Step 10: Customer downloads the exported data file.
Description of the Alternative Sequence	• Step A1: Customer is not authenticated → System requests login before proceeding. • Step A2: No data available → System generates an empty or partial export and informs the customer. • Step A3: System fails during data generation → System displays an error message and logs the failure.

	• Step A4: Export file generation takes longer than expected → System notifies the customer and provides delayed download access. • Step A5: Customer cancels the request → System terminates the process without generating the file.
Non-functional requirements	The system must ensure secure handling and transmission of personal data (encryption in transit and at rest). The system must comply with data protection regulations (e.g., GDPR). The system must generate export files within acceptable performance limits. The system must provide clear feedback and status updates to the customer.
Postconditions	• A downloadable file containing the customer's personal data is generated. • The customer successfully downloads the requested data export.

Author	Sindi Terihati
UC Name	UC-NEW-3 – Access Order Tracking and Invoice Download Using Order Reference Number and Email Address, Without a Registered Account
Summary	Allows a guest customer — who completed a purchase via guest checkout without creating an account — to access their order tracking information and download their invoice using only their order reference number and email address. This use case exists because FR-11 (Sindi Terihati) enables guest checkout, which creates customers who have no account through which to access UC-16 (tracking) or UC-18 (invoices). Based on: FR-NEW-3 – Sindi Terihati. Required by: FR-11 – Sindi Terihati.
Dependency	Depends on: • UC-11 – Guest Checkout (Sindi Terihati) — this use case exists solely because UC-11 enables guest purchasing, which creates customers who have no account through which to access tracking or invoices. • UC-48-49 – Platform Analytics & Inventory Synchronisation (Arjola Metaliaj) — the tracking data accessible to guests is sourced from the same logistics API integration (FR-49) in UC-48-49. • UC-46-47 – Order Validation & Invoice Generation (Arjola Metaliaj) — the invoice accessible to guests is the same PDF generated by FR-46 in UC-46-47. Integrates with: • UC-16 – Track Delivery (Elisa Spaho) — when a guest selects Track Delivery, this use case delegates to UC-16 for the live map and ETA display. • UC-18 – Download Invoice (Elisa Spaho) — when a guest selects Download Invoice, this use case delegates to UC-18 for the PDF retrieval and download. UC-29 – Submit Support Ticket (Ederta Kariqi) — used when guest order data is no longer accessible beyond the 90-day retention period
Actors	Primary Actor: Customer (guest — not registered) Secondary Actor: Logistics Provider – provides tracking data via API (FR-49 – Arjola Metaliaj)

Preconditions	• <i>The customer has completed a purchase via guest checkout (FR-11 – Sindi Terihati).</i> • <i>The customer is not logged into a registered SOSS account.</i> <i>The customer has access to their order reference number and the email address used at checkout.</i>
Description of the Main Sequence	1. <i>Customer navigates to 'Track Order' or 'Access My Order' from the platform homepage or order confirmation email.</i> 2. <i>System displays a guest order access form requesting: order reference number and email address.</i> 3. <i>Customer enters their order reference number and email address and submits.</i> 4. <i>System validates the combination — the email address must match the one used at the time of guest checkout.</i> 5. <i>System retrieves the order record and displays the guest order detail page, showing: order status, itemised products, delivery address, and two available actions: Track Delivery and Download Invoice.</i> 6. <i>If the customer selects Track Delivery: System queries the logistics API (FR-49 – Arjola Metaliaj) and displays the live map and ETA, fulfilling UC-16 – Elisa Spaho for guest customers.</i> 7. <i>If the customer selects Download Invoice: System retrieves the PDF invoice (FR-46 – Arjola Metaliaj) and initiates the download, fulfilling UC-18 – Elisa Spaho for guest customers.</i>
Description of the Alternative Sequence	• Alt-A — Order reference number or email address does not match: • – <i>System displays: 'We could not find an order matching those details. Please check your order confirmation email.' No order data is revealed per NFR-2 – Elda Toska.</i> • – <i>Customer may retry up to three times before the system temporarily blocks further attempts.</i> • Alt-B — Order data no longer accessible (beyond 90-day guest retention period): • – <i>System displays: 'Guest order access is available for 90 days after the order date. This order is no longer accessible. Please contact support if you require documentation.'</i> • – <i>Customer is directed to submit a support ticket (FR-29 – Ederta Kariqi).</i> • Alt-C — Logistics API unavailable during tracking attempt: • <i>System displays the last known tracking status following the same fallback behaviour as UC-16 Alt-A – Elisa Spaho.</i>
Non-functional requirements	• NFR-2 – Elda Toska: <i>The system shall protect customer personal data and purchase history from unauthorised access. Guest order access must only be granted when both the order reference number and email address match exactly — neither field alone is sufficient.</i> • NFR-11 – Elisa Spaho: <i>The system shall maintain detailed logs of user actions and transactions. All guest order access attempts, successful or failed, must be logged with timestamp and IP address.</i> • NFR-13 – Elisa Spaho: <i>The system's logistics API integration shall maintain a synchronisation latency of no more than 30 seconds and reconnect automatically after connection drops. This applies equally to guest tracking requests.</i>

	• NFR-10 – Sindi Terihati: The system shall be compatible with major web browsers and mobile devices. The guest order access page must be fully functional on mobile devices, as many guest customers will access it from a mobile order confirmation email. Guest order data shall be retrievable for a minimum of 90 days after order completion, after which it must be anonymised in compliance with GDPR data minimisation principles (NFR-17 – Stilianos Troka).
Postconditions	• The guest customer has successfully accessed their order detail page using their order reference number and email address (FR-NEW-3 – Elisa Spaho). • If Track Delivery was selected, the guest customer can see their live delivery status and ETA (UC-16 – Elisa Spaho). • If Download Invoice was selected, the guest customer has downloaded their PDF invoice (UC-18 – Elisa Spaho). The access attempt is logged for security and audit purposes.

Author	Stilianos Troka
UC Name	UC-22 Manage Payment Methods
Summary	<i>This use case allows the customer to manage multiple payment methods in their account, including adding new payment methods, editing existing ones, and deleting saved payment methods.</i>
Dependency	Includes: • User Authentication and Session Validation • Payment Method Validation • Secure Tokenization / Encryption Service Depends on: • UC-10 – Order and Checkout Management • Customer Account Management Module • Payment Gateway / Payment Processing Service Integrates with: • UC-46-47 – Order Validation & Invoice Generation • Fraud Detection and Security Monitoring Module
Actors	<i>Primary Actor: Customer</i>

Preconditions	• Customer is registered in the system. • Customer is logged into their account.
Description of the Main Sequence	• Step 1: Customer navigates to the account/payment settings section. • Step 2: Customer selects the "Manage Payment Methods" option. • Step 3: System retrieves and displays the list of saved payment methods. • Step 4: Customer selects a payment management operation (add, edit, or delete). • Step 5: System displays the appropriate interface based on the selected operation. • Step 6: Customer performs the selected action (enter new payment details, modify existing details, or confirm deletion). • Step 7: System validates the payment information or action. • Step 8: System securely processes the payment information. • Step 9: System updates the payment methods accordingly. • Step 10: System confirms successful operation. • Step 11: System refreshes and displays the updated list of payment methods.
Description of the Alternative Sequence	• Step A1: Customer enters invalid or incomplete payment details → System displays validation errors and requests correction. • Step A2: Customer cancels the operation → System returns to the payment methods list without saving changes. • Step A3: System fails during processing or storage → System displays an error message and retains the previous state. • Step A4: Customer attempts to delete a payment method currently in use for an active transaction → System blocks deletion and displays a warning.
Non-functional requirements	The system must ensure strong security by encrypting all sensitive payment data both in transit and at rest. Role-based access control must restrict access to authorized users only. The system must comply with financial and data protection regulations. Performance requirements must ensure quick validation and response times. The interface must provide clear feedback and error handling.
Postconditions	• The customer's payment methods list is updated based on the selected operation. • All valid changes are securely stored in the system. • Updated payment methods are available for future transactions.

Author	Stilianos Troka
UC Name	UC-23 Manage Delivery Addresses
Summary	<i>This use case allows the customer to manage multiple delivery addresses in their account, including adding new addresses, editing existing ones, and deleting addresses for future order use.</i>
Dependency	Includes: • User Authentication and Authorization • Address Validation Service

	Depends on: • UC-10 – Order and Checkout Management • Customer Account/Profile Management • Shipping and Delivery Configuration Module Integrates with: • UC-44-45 – Product Information Display • Delivery Tracking System • Audit Logging System
Actors	Primary Actor: Customer
Preconditions	• Customer must be registered and logged into the system. • Customer account must be active.
Description of the Main Sequence	• Step 1: Customer navigates to the account/profile section. • Step 2: Customer selects the “Manage Delivery Addresses” option. • Step 3: System retrieves and displays the list of saved delivery addresses. • Step 4: Customer selects an address management operation (add, edit, or delete). • Step 5: System displays the appropriate interface based on the selected operation. • Step 6: Customer performs the selected operation (enter new data, modify existing data, or confirm deletion). • Step 7: System validates the input or action. • Step 8: System updates the address data accordingly. • Step 9: System confirms the successful operation. • Step 10: System refreshes and displays the updated list of delivery addresses.
Description of the Alternative Sequence	• Step A1: Customer enters invalid or incomplete data → System displays validation errors and requests correction. • Step A2: Customer cancels the operation → System returns to the address list without making changes. • Step A3: System fails to process the request → System displays an error message and retains the previous state. • Step A4: No existing addresses found → System displays an empty list and prompts customer to add a new address.
Non-functional requirements	The system must enforce secure authentication and role-based access control to ensure only authorized users can manage addresses. All personal address data must be securely stored and comply with data protection regulations. The system should respond to user actions within acceptable performance limits and provide clear feedback messages. Input validation must ensure data accuracy and consistency.
Postconditions	• The customer’s delivery address list is updated based on the selected operation. • All changes are stored in the system database. • The updated address information is available for use during checkout.

Author	Stilianos Troka
UC Name	UC-24 Submit Review and Rating
Summary	<i>This use case allows the customer to submit reviews and ratings for products they have previously purchased, enabling feedback and evaluation for other users.</i>
Dependency	Includes: • View Order History • Content Validation and Moderation Depends on: • UC-10 – Order and Checkout Management • Product Catalog Management • Customer Authentication Module Integrates with: • UC-44-45 – Product Information Display • Recommendation and Ranking Engine • Notification System • Review Moderation/Admin Dashboard
Actors	<i>Primary Actor: Customer</i>
Preconditions	• Customer is registered and logged into the system. • Customer has completed the purchase of the product. • Product exists in the customer's order history.
Description of the Main Sequence	• Step 1: Customer navigates to the order history or product page. • Step 2: Customer selects a product they have purchased. • Step 3: Customer chooses the option to submit a review and rating. • Step 4: System displays a review input interface (text field and rating scale). • Step 5: Customer enters review text and selects a rating. • Step 6: Customer submits the review. • Step 7: System validates the submitted content. • Step 8: System stores the review and rating in the database. • Step 9: System associates the review with the corresponding product. • Step 10: System displays the review on the product page.
Description of the Alternative Sequence	• Step A1: Customer attempts to review a product not purchased → System blocks action and displays a message. • Step A2: Customer submits incomplete or invalid content → System displays validation errors and requests correction. • Step A3: Customer cancels submission → System exits without saving data. • Step A4: System fails during submission → System displays error message and

	<i>does not store the review.</i>
Non-functional requirements	<i>The system must ensure usability by providing a simple and intuitive interface for submitting reviews. Input validation must be performed efficiently with quick response times. The system must support content moderation and prevent inappropriate submissions. Stored data must be handled securely and consistently.</i>
Postconditions	• <i>The review and rating are stored in the system.</i> • <i>The review is associated with the corresponding product.</i> • <i>The review becomes visible to other users on the product page.</i>

Author	Stilianos Troka
UC Name	UC-25 View Personalized Product Recommendations
Summary	<i>This use case allows the customer to view personalized product recommendations generated based on their browsing activity and purchase history.</i>
Dependency	Includes: • <i>Recommendation Generation Engine</i> • <i>User Behavior Analysis</i> • <i>Preference and Activity Tracking</i> Depends on: • <i>UC-43 – Personalized Recommendation Engine (Arjola)</i> • <i>Customer Browsing History</i> • <i>Purchase History Database</i> • <i>Product Catalog Management</i> Integrates with: • <i>UC-01 – Search Products</i> • <i>UC-09 – Wishlist</i> • <i>Product Analytics Module</i>
Actors	<i>Primary Actor: Customer</i>
Preconditions	• <i>Customer is registered and logged into the system.(optional but preferred)</i> • <i>Customer has browsing history and/or purchase history available in the system.</i>

Description of the Main Sequence	• Step 1: Customer accesses the homepage or recommendation section. • Step 2: System retrieves customer browsing history, purchase data, and preferences. • Step 3: System analyzes the collected data using recommendation logic. • Step 4: System generates a list of personalized product recommendations. • Step 5: System displays the recommended products to the customer. • Step 6: Customer views and explores the recommended products.
Description of the Alternative Sequence	• Step A1: Insufficient user data available → System generates and displays generic or popular product recommendations. • Step A2: Customer is not logged in → System displays non-personalized recommendations. • Step A3: System fails to generate recommendations → System displays fallback content or an error message.
Non-functional requirements	The system must generate recommendations efficiently with minimal response time. It must scale to handle large volumes of user data and concurrent users. Recommendation accuracy should be high to ensure relevance. The system must comply with data privacy regulations when processing user behavior and preferences.
Postconditions	• Personalized product recommendations are displayed to the customer. • Customer interaction data may be updated for improving future recommendations.

Author	Ederta Kariqi
UC Name	UC-29 Submit Support Ticket
Summary	The system enables customers to submit support tickets to report issues related to orders, payments, or products.
Dependency	Depends on user account, order/payment data, and support management system. Depends on: • UC-10 – Order and Checkout Management (Sindi Terihati) • UC-22 – Manage Payment Methods (Stilianos Troka) • UC-23 – Manage Delivery Addresses (Stilianos Troka) Integrates with: • UC-15 – Submit Refund Requests (Elisa Spaho) • UC-17 – Notifications (Elisa Spaho)

Actors	<i>Primary: Customer</i> <i>Secondary: Support Team</i>
Preconditions	<i>Customer is logged into the system</i> <i>Customer has an existing order, payment, or product issue</i> <i>Support module is available</i>
Description of the Main Sequence	1. <i>Customer navigates to the Help/Support section</i> 2. <i>Customer selects "Submit Ticket"</i> 3. <i>System displays ticket form</i> 4. <i>Customer selects issue type (order/payment/product)</i> 5. <i>Customer enters details and optional attachments</i> 6. <i>Customer submits the ticket</i> 7. <i>System validates input</i> 8. <i>System creates ticket and assigns ID</i> 9. <i>System confirms submission</i>
Description of the Alternative Sequence	• <i>4a. Missing required fields → system prompts user to complete them</i> • <i>5a. Invalid attachment → system shows error</i> • <i>7a. Validation fails → ticket not submitted</i> • <i>8a. System error → user asked to retry</i>
Non-functional requirements	<i>The system must ensure high reliability and continuous availability (NFR-21). It must support graceful degradation so support submission remains functional even if some services fail (NFR-22). The interface must comply with WCAG 2.1 accessibility standards (NFR-23). Error messages must be clear and not expose internal details (NFR-24).</i>
Postconditions	• <i>Ticket is stored in system</i> • <i>Customer receives confirmation and ticket ID</i>

Author	<i>Ederta Kariqi</i>
UC Name	<i>UC-30-31 View Sustainability & Product Review Content</i>
Summary	<i>The system allows customers to view carbon footprint reports, video-based unboxing content, and product reviews before purchasing.</i>
Dependency	<i>Depends on checkout system, shipping options, and carbon calculation module.</i> <i>Depends on:</i> • <i>UC-10 – Order and Checkout Management (Sindi Terihati)</i> <i>Integrates with:</i> • <i>UC-44-45 – Product Information Display (Arjola Metaliaj)</i>

	• UC-16 – Delivery Tracking (Elisa Spaho)
Actors	Primary: Customer
Preconditions	Customer has items in cart Shipping methods are available Carbon footprint data exists Video review content is available
Description of the Main Sequence	1. Customer proceeds to checkout 2. System displays shipping options 3. Customer selects shipping method 4. System retrieves carbon footprint data 5. System displays report 6. Customer reviews sustainability and product review information
Description of the Alternative Sequence	1. 3a. No selection → report not shown 2. 4a. Data unavailable → system shows message 3. 5a. Report fails → retry option 4. 6a. Video unavailable → show text-based reviews only
Non-functional requirements	The system must provide accurate and fast performance when displaying reports. It must ensure high availability (NFR-22) and reliability (NFR-21). The interface must follow accessibility standards (WCAG) (NFR-23).
Postconditions	• Customer is informed about carbon impact • Customer continues checkout

Author	Ederta Kariqi
UC Name	UC-32-33 Manage Product Listings
Summary	The system allows sellers to add, edit, update, and remove product listings including descriptions, prices, and images.
Dependency	Depends on seller account and product management system. Depends on: • UC-50 – Seller Vendor Subscription & Activation (Arjola Metaliaj) Integrates with: • UC-34 – Create Product Bundle (Ederta Kariqi) • UC-51 – Vendor Dashboard Management (Arjola Metaliaj)

	• UC-36 – Monitor Inventory Levels (Asja Xhelezi)
Actors	Primary: Seller
Preconditions	Seller is logged in Seller has permission to add products
Description of the Main Sequence	1. Seller opens product dashboard 2. Seller selects add, edit, update, or remove product 3. Platform displays product form and listing details 4. Seller enters or updates product information 5. Seller uploads or updates images 6. Seller submits changes 7. Platform validates product data 8. Platform saves changes 9. Platform updates product listing visibility
Description of the Alternative Sequence	• 4a. Missing or invalid data → correction required • 5a. Upload fails → retry • 7a. Validation fails → product changes not saved • 8a. System error → retry
Non-functional requirements	The system must ensure data integrity and reliability (NFR-21). It must allow operation even during partial failures (NFR-22). It must be deployable across cloud platforms (NFR-25). Error messages must be clear (NFR-24).
Postconditions	• Product listings are added, updated, or removed successfully • Changes are reflected in the platform catalog

Author	Ederta Kariqi
UC Name	UC-34 Create Product Bundle
Summary	The system allows sellers to create bundles of multiple products with a combined price.
Dependency	Depends on product catalog and pricing system. Depends on: • UC-32 – Add Product Listing (Ederta Kariqi) • UC-33 – Manage Product Listings (Ederta Kariqi) Integrates with: • UC-38 – AI Price Optimization (Asja Xhelezi) • UC-41-42 – Customer Engagement Analytics (Asja Xhelezi) • UC-51 – Vendor Dashboard Management (Arjola Metaliaj)

Actors	<i>Primary: Seller</i>
Preconditions	<i>Seller is logged in Products exist</i>
Description of the Main Sequence	1. <i>Seller opens bundle section</i> 2. <i>Seller selects products</i> 3. <i>Seller sets price</i> 4. <i>Seller submits bundle</i> 5. <i>System validates</i> 6. <i>System creates bundle</i> 7. <i>System publishes bundle</i>
Description of the Alternative Sequence	• <i>2a. Not enough products → warning</i> • <i>3a. Invalid price → correction</i> • <i>5a. Validation fails → retry</i>
Non-functional requirements	<i>The system must ensure accuracy and efficiency . It must support reliable operation (NFR-21) and deployment flexibility (NFR-25) .</i>
Postconditions	• <i>Bundle created</i> • <i>Bundle available in catalog</i>

Author	<i>Ederta Kariqi</i>
UC Name	<i>UC-35 View Sales Reports & Analytics</i>
Summary	<i>The system allows sellers to view sales reports including revenue and product performance.</i>
Dependency	<i>Depends on analytics and transaction data.</i> <i>Depends on:</i> • <i>UC-48 – Platform Analytics (Arjola Metaliaj)</i> <i>Integrates with:</i> • <i>UC-36 – Monitor Inventory Levels (Asja Xhelezi)</i> • <i>UC-37 – Predict Inventory Needs (Asja Xhelezi)</i> • <i>UC-38 – AI Price Optimization (Asja Xhelezi)</i> • <i>UC-39-40 – Vendor Subscription & Dashboard Access (Asja Xhelezi)</i>
Actors	<i>Primary: Seller</i>
Preconditions	<i>Seller is logged in Sales data exists</i>

Description of the Main Sequence	1. Seller opens analytics dashboard 2. System retrieves data 3. System calculates metrics 4. System generates report 5. System displays report 6. Seller reviews data
Description of the Alternative Sequence	• 2a. No data → empty report • 4a. Report fails → retry
Non-functional requirements	The system must provide high performance and fast report generation . It must ensure reliability (NFR-21) and availability (NFR-22) . The interface must be clear and accessible (NFR-23) .
Postconditions	• Seller views analytics • Seller gains business insights

Author	Asja Xhelezi
UC Name	UC-36 Monitor Inventory Levels
Summary	The seller monitors stock levels and receives alerts when inventory reaches low-stock thresholds.
Dependency	Depends on inventory database and notification services. Depends on:

	• UC-33 – Manage Products <i>Integrates with:</i> • UC-20 – Reorder • UC-08 – Shopping Cart
Actors	Primary: Seller Secondary: None
Preconditions	• Seller is logged in • Products exist in inventory
Description of the Main Sequence	1. Seller accesses the inventory dashboard 2. Platform retrieves current stock data 3. Platform displays inventory levels 4. Platform evaluates low-stock thresholds 5. Platform sends low-stock alerts to the seller
Description of the Alternative Sequence	• 2a. If no inventory data exists, the platform displays an empty inventory list • 5a. If notifications fail, alerts are displayed in the dashboard only
Non-functional requirements	Inventory data should update in near real time, and low-stock alerts should be delivered with minimal delay.
Postconditions	• Seller is informed about inventory status and low-stock alerts

Author	Asja Xhelezi
UC Name	UC-37 Predict Inventory Needs
Summary	The system analyzes historical sales data to forecast restocking needs.
Dependency	Depends on analytics services and sales databases. Depends on: • UC-35 – View Sales Reports & Analytics integrates with: • UC-36 – Monitor Inventory Levels • UC-33 – Manage Products
Actors	Primary: Seller Secondary: None

Preconditions	<i>Seller is logged in Sales history exists</i>
Description of the Main Sequence	1. Seller requests inventory analytics 2. Platform retrieves historical sales and inventory data 3. Platform analyzes sales trends and inventory movement 4. Platform predicts future stock demand 5. Platform generates restocking recommendations 6. Seller reviews inventory predictions
Description of the Alternative Sequence	• 2a. If no historical data exists, the platform provides basic inventory suggestions • 3a. If analysis processing fails, the platform displays fallback prediction results
Non-functional requirements	<i>Inventory predictions should be generated efficiently and provide accurate forecasting insights for seller restocking decisions.</i>
Postconditions	<i>Seller receives insights for restocking decisions</i>

Author	<i>Asja Xhelezi</i>
UC Name	<i>UC-38 AI Price Optimization</i>
Summary	<i>The system adjusts product prices automatically using AI for better market positioning.</i>
Dependency	<i>Depends on pricing rules and product management services.</i> <i>Depends on:</i> • UC-33 – Manage Products <i>Integrates with:</i> • UC-36 – Monitor Inventory Levels • UC-35 – View Sales Reports & Analytics
Actors	<i>Primary: Seller</i> <i>Secondary: None</i>
Preconditions	<i>Seller is logged in Product and market data exist</i>
Description of the	<i>1. Seller accesses dynamic pricing tools</i>

Main Sequence	2. Platform retrieves product, inventory, and sales data 3. Platform analyzes pricing conditions and sales performance 4. Platform generates optimized pricing suggestions 5. Seller reviews and applies pricing updates 6. Platform updates product pricing
Description of the Alternative Sequence	• 2a. If pricing data is unavailable, the platform displays the current product prices only • 4a. If pricing analysis fails, the platform provides default pricing suggestions
Non-functional requirements	Pricing recommendations should be generated quickly and reflect current inventory levels, product demand, and sales trends accurately.
Postconditions	Product prices are optimized automatically

Author	Asja Xhelezi
UC Name	UC-39-40 Vendor Subscription & Dashboard Access
Summary	The seller subscribes to vendor plans, renews subscriptions, and gains access to vendor dashboard functionalities.
Dependency	Depends on seller subscription services and payment processing services. Depends on: • UC-22 – Payment Methods Integrates with: • UC-50 – Seller Vendor Subscription & Activation • UC-51 – Vendor Dashboard Management
Actors	Primary: Seller Secondary: Payment Gateway
Preconditions	Seller is logged in Seller account exists
Description of the Main Sequence	1. Seller accesses subscription management 2. Platform displays available vendor plans and subscription details 3. Seller selects or renews a subscription plan 4. Platform redirects seller to payment processing 5. Payment gateway processes the payment 6. Platform validates the payment result 7. Platform activates or renews the seller subscription 8. Seller gains continued access to vendor features and dashboard tools

Description of the Alternative Sequence	• 5a. If payment fails, the seller is informed and may retry payment • 6a. If payment validation fails, the subscription is not activated or renewed • 3a. If the seller cancels the process, no subscription changes are applied
Non-functional requirements	Subscription processing should be secure, reliable, and completed with minimal delay so sellers can quickly access vendor platform features.
Postconditions	Seller subscription is activated or renewed successfully Seller gains access to vendor dashboard functionalities

Author	Asja Xhelezi
UC Name	UC-41-42 Customer Engagement Analytics
Summary	The seller analyzes customer engagement insights, reviews, and recommendation analytics.
Dependency	Depends on analytics services, customer feedback data, and recommendation services. Depends on: • UC-35 – View Sales Reports & Analytics Integrates with: • UC-25 – Personalized Product Recommendations • UC-24 – Submit Review • UC-44-45 – Product Information Display
Actors	Primary: Seller Secondary: None
Preconditions	Seller is logged in Customer engagement data exists
Description of the Main Sequence	1. Seller accesses customer engagement analytics 2. Platform retrieves customer behavior, review, and recommendation data 3. Platform analyzes customer interaction and product engagement 4. Platform generates engagement insights and recommendations 5. Seller reviews engagement analytics and optimization suggestions
Description of the Alternative	• 2a. If customer engagement data is unavailable, the platform displays limited analytics information

Sequence	• 4a. If recommendation analysis fails, the platform displays basic engagement statistics only
Non-functional requirements	Customer engagement analytics should process data efficiently and provide accurate insights based on customer interactions, reviews, and recommendation activity.
Postconditions	Seller receives customer engagement insights and recommendation analytics

Author	Arjola Metaliaj
UC Name	UC-44-45 Product Information Display
Summary	The customer views detailed product information, including product description, reviews, ratings, and estimated delivery time.
Dependency	Depends on: • UC-24 – Submit Review Integrates with: • UC-16 – Delivery Tracking.
Actors	Primary: Customer
Preconditions	• Product exists in the marketplace catalog.
Description of the Main Sequence	1. Customer selects a product. 2. Platform retrieves product details. 3. Platform displays product description, price, images, and availability. 4. Platform displays customer reviews and ratings. 5. Platform calculates estimated delivery time. 6. Platform displays the delivery estimate to the customer.
Description of the Alternative Sequence	• 4a. If no reviews are available, the platform displays “No reviews yet.” • 5a. If delivery estimation is unavailable, the platform informs the customer that the delivery estimate cannot be shown.
Non-functional requirements	Product pages should load quickly, including reviews and delivery estimates. Product information should be highly available and consistently accessible. Reviews, ratings, and delivery estimates should be accurate and reliable so customers can make informed purchasing decisions.
Postconditions	• Customer is informed about product quality, product details, and delivery expectations.

Author	Arjola Metaliaj
UC Name	UC-46-47 Order Validation & Invoice Generation
Summary	<i>The customer's order details are validated before confirmation and they receive a downloadable invoice after successful purchase.</i>
Dependency	Depends on: • UC-10 – Checkout Integrates with: • UC-18 – Invoice Download • UC-22 – Payment Methods
Actors	Primary: Customer Secondary: Payment Gateway
Preconditions	• Customer has items in cart • Checkout process initiated
Description of the Main Sequence	1. Customer proceeds to checkout. 2. Platform validates payment information, stock availability, and delivery address. 3. Platform sends payment request to the payment gateway. 4. Payment gateway confirms successful payment. 5. Platform confirms the order. 6. Platform generates the invoice or receipt. 7. Customer downloads the invoice or receipt.
Description of the Alternative Sequence	• 2a. If payment information is invalid, the platform rejects the transaction and asks the customer to correct it. • 2b. If a product is out of stock, the platform notifies the customer and prevents checkout confirmation. • 2c. If the delivery address is invalid, the platform asks the customer to update the address. • 4a. If payment fails, the platform cancels the order confirmation and informs the customer. • 6a. If invoice generation fails, the platform logs the issue and allows the customer to request the invoice again.
Non-functional requirements	<i>Order validation must be secure, with payment and order data protected against unauthorized access. Validation and confirmation should be completed quickly to support a smooth checkout experience. Invoice generation should be reliable, accurate, and stored without data loss.</i>
Postconditions	• Order successfully recorded • Invoice generated and available for download

Author	Arjola Metaliaj
UC Name	UC-48-49 Platform Analytics
Summary	<i>The administrator views platform analytics based on sales, usage, customer activity, and seller performance data.</i>
Dependency	Depends on: • <i>UC-10 – Checkout because completed orders and transactions provide sales data for analytics.</i>
Actors	<i>Primary: Administrator</i>
Preconditions	• <i>Administrator is logged in.</i> • <i>Sales, usage, customer activity, or seller performance data exists.</i>
Description of the Main Sequence	1. <i>Administrator opens the analytics dashboard.</i> 2. <i>Platform retrieves sales, usage, customer activity, and seller performance data.</i> 3. <i>Platform processes the collected data.</i> 4. <i>Platform generates analytics reports, charts, and summaries.</i> 5. <i>Administrator views the platform analytics.</i>
Description of the Alternative Sequence	• <i>2a. If no data is available, the platform displays an empty analytics dashboard.</i> • <i>3a. If analytics processing fails, the platform displays an error message and logs the issue.</i>
Non-functional requirements	<i>Analytics should load quickly and present accurate, up-to-date information. The feature should support reliable data processing and maintain data consistency across sales, usage, customer activity, and seller performance records.</i>
Postconditions	• <i>Analytics available to administrators</i>

Author	Arjola Metaliaj
UC Name	UC-50 Seller Vendor Subscription & Activation
Summary	<i>The seller subscribes to a vendor plan and becomes an active vendor after successful payment validation.</i>
Dependency	Depends on: • <i>UC-22 – Payment Methods</i>

	<i>Integrates with:</i> • <i>UC-32 – Add Products (Seller)</i> • <i>UC-33 – Manage Products (Seller)</i> • <i>UC-51 – Vendor Dashboard Management because these features become available after vendor activation.</i>
Actors	<i>Primary: Seller</i> <i>Secondary: Payment Gateway</i>
Preconditions	• <i>Seller account exists.</i> • <i>Seller is logged in.</i> • <i>Seller is not yet subscribed to a vendor plan.</i>
Description of the Main Sequence	1. <i>Seller selects “Become a Vendor.”</i> 2. <i>Platform displays available vendor subscription plans.</i> 3. <i>Seller selects a subscription plan.</i> 4. <i>Platform redirects the seller to payment processing.</i> 5. <i>Seller completes payment through the payment gateway.</i> 6. <i>Payment gateway returns the payment result.</i> 7. <i>Platform validates the successful payment.</i> 8. <i>Platform activates the seller’s vendor account.</i> 9. <i>Seller gains access to product listing and selling features.</i>
Description of the Alternative Sequence	• <i>5a. If payment fails, the seller is informed and may retry the payment.</i> • <i>5b. If the seller cancels payment, the subscription process terminates and the account remains inactive.</i> • <i>7a. If payment validation fails, the vendor account is not activated.</i>
Non-functional requirements	<i>Payment processing must be secure and protect sensitive financial data. Vendor activation should be reliable and highly available. After successful payment validation, activation should happen quickly so the seller can access selling features without delay.</i>
Postconditions	• <i>Seller becomes an active vendor</i> • <i>Seller can access selling functionalities such as adding and managing products</i>

Author	<i>Arjola Metaliaj</i>
UC Name	<i>UC-51 Vendor Dashboard Management</i>
Summary	<i>The seller manages the vendor dashboard to monitor sales performance analytics, product listings, and synchronized inventory information.</i>
Dependency	<i>Depends on:</i> • <i>UC-50 – Seller Vendor Subscription & Activation</i>

	<i>Integrates with:</i> • <i>UC-32 – Add Products</i> • <i>UC-33 – Manage Products</i> • <i>UC-36 – Inventory Monitoring</i> • <i>UC-48 – Platform analytics</i>
Actors	<i>Primary: Seller</i>
Preconditions	• <i>Seller is logged in.</i> • <i>Seller has an active vendor account.</i>
Description of the Main Sequence	1. <i>Seller opens the vendor dashboard.</i> 2. <i>Platform retrieves seller performance data and product information.</i> 3. <i>Platform displays sales analytics and performance summaries.</i> 4. <i>Platform displays product listings managed by the seller.</i> 5. <i>Platform displays synchronized inventory information.</i> 6. <i>Seller reviews and manages dashboard information.</i>
Description of the Alternative Sequence	• <i>2a. If analytics data is unavailable, the platform displays limited dashboard information.</i> • <i>5a. If inventory synchronization fails, the platform displays the most recent available inventory data and logs the issue.</i>
Non-functional requirements	<i>Dashboard data should load quickly and update in near real time. Analytics and inventory information should remain accurate and consistent across seller and customer views. The dashboard should maintain high availability and usability even with large amounts of seller data.</i>
Postconditions	• <i>Seller can monitor sales performance, manage listings, and review inventory information through the dashboard.</i>